

OMOUAI: Argumentative Human-AI Policy Deliberation with Simulated Personas

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Abstract

Debates amongst agents driven by large language models (LLMs) have demonstrated vast potential in various applications, but when these interactions include humans and take place in high-stakes environments, e.g., in public policy deliberations, they are beset with issues such as sycophancy and a lack of faithful explanations. To tackle these issues, we present OMOUAI, an interactive and inclusive deliberation system that uses LLMs in combination with computational argumentation, a field which excels in representing and reasoning within debates. OMOUAI allows a human user to deliberate policy claims for real-world challenges with simulated personas, e.g., representing stakeholders, domain experts or devil’s advocates, towards reducing sycophancy. Each persona generates its own arguments, and the arguments of all parties form a shared argumentation framework. Users can then contest, add and revise arguments, providing crucial human oversight. Then, arguments are evaluated using deterministic argumentative semantics against external goals, such as the UN Sustainable Development Goals, guaranteeing faithful explanations. The advancement or worsening of the goals thus serve as indicators for the policy recommendations.

Introduction

AI agents built on large language models (LLMs) now support decisions in high-stake domains such as medicine, law, and public administration (Wang et al. 2024). These agents have been shown to be particularly powerful when their decision making is the result of debate (Khan et al. 2024). Indeed, the Habermas Machine (Tessler et al. 2024) showed that LLMs can help citizens find common ground on divisive policy questions. Policy deliberation is a natural setting for such agents, since policy questions are contested by parties who hold different values and weigh the same evidence differently. These agents, however, often generate fluent text and reason over unstructured inputs without clearly exposing their reasoning, so their outputs cannot be faithfully explained or reliably contested. In high-stakes deliberations, reasoning that cannot be explained, evaluated or contested is of limited use. Further, LLMs are known to display “sycophantic” behaviour towards their users (Bo et al. 2026), which has no place in effective deliberation.

Meanwhile, computational argumentation (Dung 1995) is an area which has shown great suitability for representing and reasoning with debates, whether between machines (Gorur,

Rago, and Toni 2026), humans (Lawrence, Visser, and Reed 2023; Ruiz-Dolz, Heras, and García-Fornes 2023; Irwin, Rago, and Toni 2022; Goffredo et al. 2025) or a combination thereof (Rago, Li, and Toni 2023). Recently, it has been proposed that argumentative mechanisms facilitate the contestation of LLM recommendations through formal frameworks for transparent, verifiable reasoning with well-defined semantics (Vasileiou et al. 2026). Indeed, recent works (Freedman et al. 2025; Zhu et al. 2025; Dejl, Gorur, and Toni 2026) allow for LLMs to generate and weigh arguments for and against a claim, then compute the verdict with gradual semantics over the resulting argumentation framework. However, such systems have not been built specifically for public policy deliberation. In turn, deliberation systems address the opposite problem: the Habermas Machine and related systems (Tessler et al. 2024; Ma et al. 2025) aggregate many human views into a statement, but they treat LLMs as “black box” mediators, leaving the underlying reasoning, and thus the justifications for decisions, unexposed.

We present OMOUAI,¹ to the best of our knowledge the first system for collaborative human-AI decision-making which combines LLMs and argumentation specifically for public policy deliberation. OMOUAI allows for: (i) an inclusive deliberative system for real-world challenges, where the human discusses with a diversity of personas (stakeholders, domain experts, or devil’s advocates); (ii) a joint, explainable and contestable human-agent argumentative thread, where arguments are generated and scored based on the personas’ specifications; and (iii) impact evaluation of the proposed arguments towards real-world goals, e.g., the United Nations Sustainable Development Goals.

System Overview

OMOUAI is a neurosymbolic system comprising an LLM layer and an argumentation layer. Before a deliberation starts, the user enters a policy claim, chooses the personas that take part, and selects indicators representing goals against which the deliberation is to be measured. During the deliberation, the LLM layer generates arguments (α_p) and scores them with a base score (τ), i.e., an intrinsic strength, on behalf

¹OMOUAI is derived from the word *omou*, which means *deliberating* in Japanese and *together* in Greek. Taken together, OMOUAI means *deliberating together with AI*.

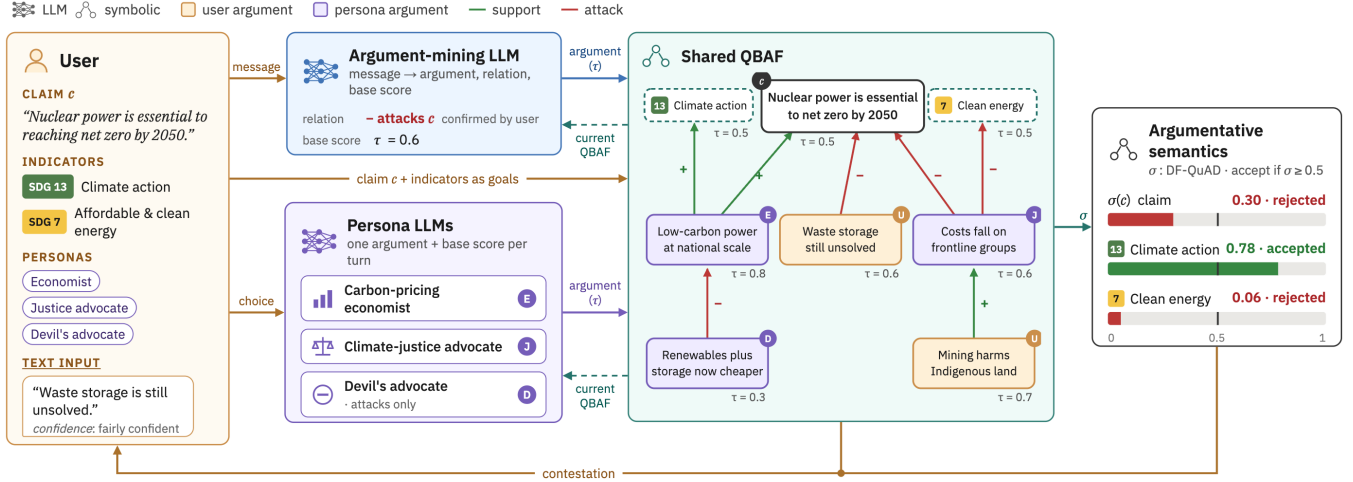


Figure 1: The OMOUAI pipeline for a climate change policy example.

of each persona, and mines the user’s messages into arguments (α_u) and base scores. In the argumentation layer, these two sets of arguments are combined in a shared quantitative bipolar argumentation framework (QBAF) (Baroni, Rago, and Toni 2018) with arguments representing the claim c and each selected indicator (α_i), all of which are evaluated with a gradual semantics (σ). Figure 1 shows the pipeline.

QBAs. The shared QBAF is a tuple $\langle \mathcal{A}, \mathcal{R}^-, \mathcal{R}^+, \tau \rangle$ comprising a set of arguments $\mathcal{A} = \{c\} \cup \alpha_p \cup \alpha_u \cup \alpha_i$, disjoint attack and support relations $\mathcal{R}^-, \mathcal{R}^+ \subseteq \mathcal{A} \times \mathcal{A}$, and a base score function $\tau : \mathcal{A} \rightarrow [0, 1]$. A gradual semantics $\sigma : \mathcal{A} \rightarrow [0, 1]$ assigns each argument a strength computed from its base score and the strengths of its attackers and supporters. OMOUAI implements several such semantics, including DF-QuAD (Rago et al. 2016) and the quadratic energy model (Potyka 2018), and the user can switch between them at runtime. The QBAF is a merged set of trees rooted at the policy claim c and the indicators. Every argument in OMOUAI additionally carries a type, which is empirical for a statement about the present state of affairs, predictive for a statement about what the policy would cause, or normative for a statement about what ought to be done.

Personas. A persona is a specification of three to five value commitments, a specification stating the kinds of evidence the persona accepts, and an epistemic disposition stating how cautiously it reasons from that evidence (following value-belief-norm theory (Stern 2000; Long et al. 2025)). These specifications are the only persona-specific content given to the LLM, which are minimised since persona prompting degrades when irrelevant attributes are added (Park et al. 2024; Hu and Collier 2024). The user selects any number of personas to take part in the deliberation. A persona can be a stakeholder, which represents an affected party and whose values condition both generation and scoring; a domain expert, which argues only from evidence; or a devil’s advocate, which only attacks and serves to test arguments.

Argument generation and scoring (α_p). Given a target argument and the arguments already attached to it, α_p returns

a single new attacker or supporter of the target on behalf of the persona, or declines if the persona has nothing to add. τ then assigns the base score according to the argument’s type, scoring an empirical or predictive argument by the evidential support the persona finds for it and a normative argument by the degree to which the persona’s commitments endorse it. The claim and indicator arguments start with a base score of 0.5, indicating a neutral stance.

Argument mining (α_u). The user’s own contributions are provided as free-text messages, and each message is first classified (by an LLM) as an argument, a request or a question (Chen et al. 2024). An argument is a message that takes a position on the claim or on an existing argument. For such a message, OMOUAI returns a restatement as a single argument, and a proposed attack or support relation to an existing argument, and the user confirms or redirects the proposed relation before the argument enters the QBAF. The user’s argument is also assigned a base score τ based on the confidence the user stated for it on a five-level scale. A request is a message that asks a persona for an argument, which results in a new argument in α_p . A question is a message that asks about the current state of the deliberation, and it is answered from the computed record of the argumentation layer.

Evaluation (σ). The argumentation layer computes a strength for all arguments, including those representing the claim and indicators, and thus the deliberation results in a verdict for each goal alongside the verdict on the claim. The implemented semantics satisfy the base score and relation contestability properties in (Freedman et al. 2025), which guarantee intuitive behaviour such as a user’s intervention always moving the verdict in the direction the user intends.

Responsible Use. OMOUAI is a human-AI argumentation-support tool, not a prescriptive decision-making system. The generated arguments reflect the biases of the underlying models and the value systems of the represented stakeholders. Domain experts must evaluate the outputs before these are considered for real-world decision making purposes.

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